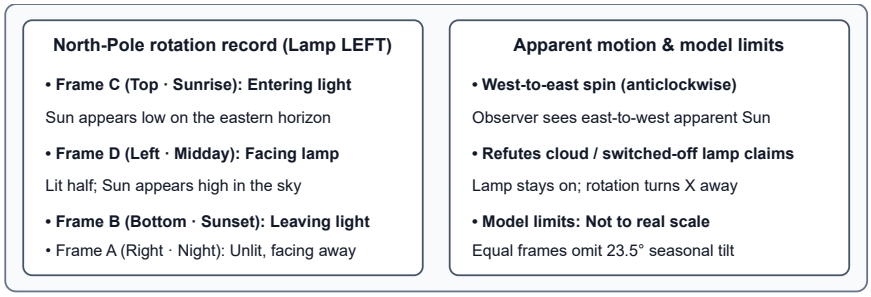
# Knowledge organiser




Knowledge organiser model. It links Frames C, D, B, and A to apparent Sun motion and model limits.

**Learning goal.** I can use a rotating-Earth model to explain day, night, and apparent Sun motion.

## Tier 2 and Tier 3 vocabulary

| Tier. | Word. | Meaning. | Example in this lesson. |
|---|---|---|---|
| Tier 3 | rotation | One full turn of Earth on its axis in 24 hours. | Earth's rotation carries our place from daylight into darkness every day. |
| Tier 3 | axis | An imaginary line from pole to pole through a turning planet. | Earth spins anticlockwise around its axis when viewed above the North Pole. |
| Tier 2 | apparent | How the Sun's movement seems from an observer turning with Earth. | Earth's west-to-east spin causes the Sun's apparent east-to-west sky motion. |
| Tier 2 | cycle | A repeating sequence of stages that returns to its starting position. | Four quarter turns return marker X to the start of the cycle. |
| Tier 2 | evidence | Recorded observations from a model or test that support a science explanation. | Our four-frame light record is evidence that rotation causes day and night. |
| Tier 2 | limit | What a simple classroom model cannot show accurately, such as real scale. | One limit of equal frames is omitting seasonal daylight changes. |

**Core Key Stage 3 science ideas.** Earth completes one anticlockwise **rotation** on its **axis** every 24 hours. We turn west to east with Earth. This gives the Sun an **apparent** east-to-west motion across

the sky. Model **evidence** (a fixed lamp and no clouds) proves rotation causes night. Honest **limits** note that classroom models are not to scale. Equal frames also simplify seasonal axial tilt.

**Made by Matt**